



Impingement and mixing between two shear-thinning droplets on the solid surface

Ziqi Li^{a,b}, Guina Yi^{a,b}, Ziqi Cai^{a,b,*}, Zhengming Gao^{a,b,*}

^a Beijing Advanced Innovation Center for Soft Matter Science and Engineering, Beijing University of Chemical Technology, Beijing 100029, China

^b State Key Laboratory of Chemical Resource Engineering, School of Chemical Engineering, Beijing University of Chemical Technology, Beijing 100029, China

HIGHLIGHTS

- Shear thinning non-Newtonian droplets are constructed by MDPD method.
- Diffusion process in non-Newtonian droplets differs from that in Newtonian droplets.
- A $We-t_c$ relationship is proposed for the control of droplets mixing on solid surface.
- Relationship between total mixing time and Weber number follows the power law.

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ABSTRACT

The impingement and the following mixing between non-Newtonian droplets on the solid surface are a common phenomenon in industry, where reasonably controlling the mixing is always of vital importance. In this work, Many Body Dissipative Particle Dynamics method is used to reveal the law of this process with two non-Newtonian droplets. Firstly, the variation of droplets morphology with the Weber number of the impinging droplet is studied. Then, by dividing this mixing process into convection-dominated and molecular diffusion-dominated stages, the relationship between duration of the former stage and Weber number is quantitatively determined. In the latter stage, it is found a special phenomenon in the mixing which results from the difference in particle diffusion coefficients. Finally, the mixing time for the two droplets and its variation with the Weber number is proposed, which can be helpful in the design of the related device when mixing is between non-Newtonian droplets.

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1. Introduction

Impingement and the following mixing between droplets on the solid surface are a common phenomenon in nature (Hu et al., 2021) and industrial processes, such as pesticide spraying (Bergeron et al., 2000), spray coating (Aziz and Chandra, 2000) and spray printing (Son and Kim, 2009). Therefore, the understanding of this phenomenon is of great value (Castrejón-Pita et al., 2013) to many applications such as coatings for dew collection, plug-based microfluidic system, and micro-batteries (Sellier and Trelluyer, 2009; Song et al., 2006; Yang et al., 2012).

The research on droplets impingement and mixing involves various topics (Xia et al., 2022) including the physical properties of droplets and the solid surface, and impinging conditions, among which the droplet is always Newtonian. For example, Mao et al.

(1997) carried out experiments into the maximum spread and rebound of liquid droplets upon impact on flat surface at room temperature, founding that liquid viscosity and impact velocity were the two of most important parameters affecting the maximum extent to which a droplet spreads upon impact. Meanwhile, viscosity and static contact angle play a dominant role in determining the tendency of a droplet to rebound after impact.

Šikalo et al. (2002) conducted experiments where droplets (water, isopropanol, glycerol) hit solid walls (paraffin, rough glass, smooth glass, and PVC). It was found that the wettability of the solid surface has a great influence on the droplet spreading, and the contact angle is important to the droplet spreading at low Reynolds number and low Weber number.

Compared with Newtonian fluid in the process industry, non-Newtonian fluid is more applied, whose shear-thinning characteristic should be always concerned. The impingement and mixing process of non-Newtonian droplets can be different from that of the Newtonian droplets due to its shear dependent property. There are many studies on the impact of a single non-Newtonian droplet

* Corresponding authors.

E-mail addresses: caiziqi@mail.buct.edu.cn (Z. Cai), gaozm@mail.buct.edu.cn (Z. Gao).

on the solid surface, while only a few is about the impingement between two non-Newtonian droplets.

Crooks et al. (2001) photographed the morphology of droplets (water, glycerol, polyethylene oxide solution, etc.) after hitting the paraffin film, and found that compared with the Newtonian fluid, an increase in the molecular mass or concentration of the polymer in solution inhibits the rebound of the droplet on the hydrophobic wall.

An and Lee (2012a, 2012b) and Ravi et al. (2013) studied the behavior of droplets of pseudoplastic fluid (xanthan gum aqueous solution) and hydroxyethyl cellulose aqueous solution hitting the wall, respectively, and pointed out that the shear-thinning property has a significant effect on the wall-impacting behavior of droplets. Since the high shear rate during impact reduces the shear viscosity of the droplet, the droplet has less viscous dissipation during spreading. Compared to Newtonian droplets with the same low shear rate viscosity, the shape evolution of a shear-thinning droplet after hitting the wall is faster and more dramatic.

Although the experimental research on the droplet-wall impingement has been widely carried out in the scale of millimeter and sub-millimeter, it is still lack of understanding of the behavior of non-Newtonian droplet in micrometer scale, due to the limitations of experimental measurement of the velocity and pressure distribution, even the concentration field during the impingement in such scale.

Numerical simulation is a powerful tool in describing the micro-droplet (in micro size) behavior, especially when the experimental measurement can be difficult. In tracking the droplet impact dynamics on solid surface, phase field, level set, VOF, LBM and DPD methods are all effective ways.

Generally, VOF method can be regarded as the most popular method in the description of interface between phases, and it can be used to calculate the complex phase interface change process conveniently with a high sharp degree of the interface sharpness and excellent volume conservation (Díaz and Ortega, 2013; Farhangi et al., 2012). However, because of the discontinuity of VOF function at the interface, it is insufficient to calculate the spatial gradient, and the tracking result of the moving interface becomes fuzzy (Sun et al., 2019). This method is also highly dependent on the grid size and quality, which limits its application in the simulation of multiple particle system or particle swarm.

Level set method is a typical method for implicit description of moving front, which has been popularly used in the simulation of gas-liquid and liquid-liquid interface in multiphase flow (Yokoi et al., 2009). Because of the continuity and smoothness of the functions, the precise spatial gradients can be obtained, but keeping the volume conservation is always a problem in this method, which may cause mass non-conservation due to the numerical dissipation and numerical error (Wong et al., 2017). In addition, the computation is time-consuming and the interface is not as sharp as the VOF method.

Phase-field method can naturally describe the wetting behavior of a droplet on a surface based on thermodynamic theory (Mino et al., 2016; Shardt et al., 2014) and be valid in solving the problem with sharp interfaces (Hassan and Wang, 2021). However, the use of the finite difference method or the finite element method in the phase field method makes too large computation, and the phase field method also limits the development of three-dimensional simulation of microstructure. The appearance of mesoscopic interface makes the phase-field equation very rigid, and it is also difficult to simulate the real time scale, which makes it hard to quantitatively fit the experiment well.

In the mesoscopic scale of numerical simulation, LBM (Lattice Boltzmann Method) (Mukherjee and Abraham, 2007; Tanaka et al., 2011; Yao et al., 2012) and DPD (Dissipative Particle Dynam-

ics) are both effective methods. DPD (Español and Warren, 2017; Feng et al., 2020; Kim and Phillips, 2004) is a meshless simulation method proposed by Hoogerbrugge and Koelman (1992), having the characteristics of MD method and LBM method. Compared with MD method, DPD method introduces dissipative force and random force to compensate for the reduction of degrees of freedom due to coarse graining, making it a powerful tool for solving problems over a wide range of length and time scales (Reddy and Abraham, 2009). Compared with LBM method, DPD method inherits the characteristics of LBM method in describing the fluid behavior, and improves the defect of lattice model used in LBM in simulating polymer fluid (Chen et al., 2006).

Many Body Dissipative Particle Dynamics (Warren, 2003) (MDPD) is evolved from DPD (Hoogerbrugge and Koelman, 1992). In DPD, the equation of state (EOS) derived from the linear conservative force has no higher-order terms of density, which makes it impossible to describe the systems with free interfaces, such as gas-liquid systems. Warren (2003) improved the traditional DPD method so that it can be used to describe the free interfaces at mesoscale, such as droplet motion on flat plates (Chen et al., 2020; Chu et al., 2021; Jin and Wang, 2021; Liang et al., 2017; Wang et al., 2019), fluid flow in capillaries (Chakrapani and Otter, 2020; Chen et al., 2012; Cupelli et al., 2008), porous media (Xia et al., 2017), and bubble motion (Wu et al., 2017; Xiao et al., 2021).

At present, although there is plenty of work on droplet collision with solid wall, the quantitative analysis on impingement of two shear thinning micro-droplets on solid wall is still insufficient up to now. In addition, controlling the morphology of the merged droplet after the impingement and the mixing time between impinging droplets is important to the resulting performance of droplet. For example, in the spray coating (Mahulkar et al., 2015) and color control in ink-jet printing (Sun et al., 2015), if the droplets are broken, the phenomenon of ink flying is prone to occur during inkjet printing, leading to a decrease in the quality of ink droplet formation and image accuracy. Also, the long mixing time will reduce the printing efficiency, and the paint not well mixed before drying will affect the color quality of the print. In this work, the MDPD method is used to study the impingement and mixing between two shear-thinning non-Newtonian droplets on the solid surface, in which these two droplets are respectively named impinging droplet and sessile droplet. Firstly, the non-Newtonian fluids involved in this paper are verified by the rheological curves. The dynamic behaviors of impinging and sessile droplets are studied by dividing the mixing into two stages, and following the influence of droplet Weber number on the mixing time is discussed.

2. Simulation method

Fig. 1 shows the schematic diagram of a droplet impinging a sessile droplet on the solid surface. The impinging droplet with the radius of R_c locates just above the sessile droplet, and then it falls and impinges on the sessile droplet with an initial velocity U_0 . When the sessile droplet lies on the solid surface, there is a contact angle θ_c . In our previous work (Yi et al., 2021), we discussed the influence of the impingement angle and the initial location on the impact of Newton droplets, and found that the impingement angle can affect the motion of droplets on the solid surface. Therefore, the diffusion distance of the particle in the diffusion dominant stage will be thereby changed, resulting the extension of mixing time, which is not conducive to the mixing. Based on the research results mentioned above, we only consider the vertical downwards impingement between droplets without considering the influence of impingement angle in this work.

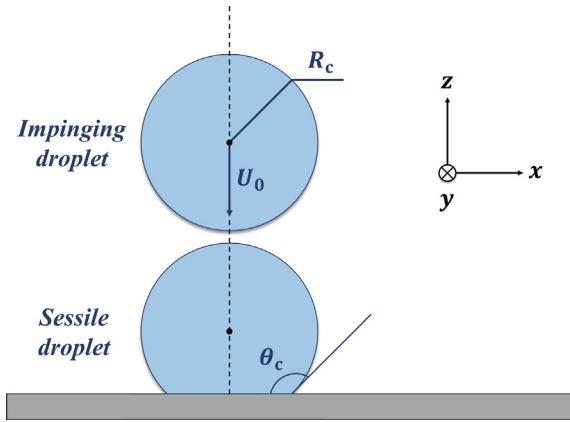


Fig. 1. Schematic diagram of a droplet impinging a sessile droplet on the solid surface.

2.1. Principle of many body dissipative particle dynamics

In terms of mesoscale research, MDPD (Du et al., 2022) is effective in simulating droplet dynamics. It can maintain a stable gas-liquid interface and can easily adjust the properties of the fluid by manipulating the interaction force.

In MDPD, the interaction between particles follows Newton's second law, and the force acting on particle includes (Wen et al., 2022): conservative force (F_{ij}^C), dissipative force (F_{ij}^D), and random force (F_{ij}^R), as shown in Eq. (1).

$$F_{ij}^{total} = F_{ij}^C + F_{ij}^D + F_{ij}^R \quad (1)$$

The expressions of the random force and dissipative force are the same as those of traditional DPD, and the attractive term is introduced into the conservative force, as follows:

$$F_{ij}^C = [A_{ij}w_c(r_{ij}) + B_{ij}(\bar{\rho}_i + \bar{\rho}_j)w_d(r_{ij})]e_{ij} \quad (2)$$

In Eq. (2), A_{ij} and B_{ij} respectively refer to the attraction coefficient and repulsion coefficient between particles, generally $A_{ij} < 0$ and $B_{ij} > 0$. r_{ij} is the distance between the particle i and the particle j , and e_{ij} is the unit vector of r_{ij} . The weight functions of attraction and repulsion, $w_c(r_{ij})$ and $w_d(r_{ij})$, can be calculated by:

$$w_c(r_{ij}) = 1 - \frac{r_{ij}}{r_c} \quad (3)$$

$$w_d(r_{ij}) = 1 - \frac{r_{ij}}{r_d} \quad (4)$$

where r_c and r_d are the cut-off radius of the attractive and repulsive interactions between MDPD particles, i.e., when $r > r_c$ the attractive force vanishes, and when $r > r_d$ the repulsive force vanishes, specifically $r_d < r_c$. And $\bar{\rho}_i, \bar{\rho}_j$ are the local densities of particle i and j , defined as $\bar{\rho}_i = \sum_{j \neq i} w_p(r_{ij})$, in which the $w_p(r_{ij})$ is calculated by $w_p(r_{ij}) = \frac{15}{2\pi^3} (1 - \frac{r_{ij}}{r_d})$.

The equations of dissipative force (F_{ij}^D) and random force (F_{ij}^R) are as follows:

$$F_{ij}^D = -\gamma w_D(r_{ij})(e_{ij} \cdot v_{ij})e_{ij} \quad (5)$$

$$F_{ij}^R = \delta w_R(r_{ij})\xi \Delta t^{-1/2} e_{ij} \quad (6)$$

where $v_{ij} = v_i - v_j$, that is, the velocity difference of the particles i and j . Here γ and δ are two coefficients representing the intensity of dissipative and random forces. To satisfy the fluctuation-dissipation theorem (Español and Warren, 1995), $\delta^2 = 2\gamma k_B T$ and

$w_D(r_{ij}) = [w_R(r_{ij})]^2$ are required. ξ is the Gaussian white noise term. Iterative calculations are performed using the standard Velocity-Verlet algorithm.

In this paper, Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS, 15 May 2019) (Plimpton, 1995) is used for the simulation, and the graphical interface is displayed by OVITO (Ver. 3.4.4). Afterwards, the analysis of output data is performed on MATLAB(R2021a).

2.2. Construction of droplets and verification of its shear-thinning property

In constructing the shear thinning non-Newtonian droplet in MDPD, we refer to the work of Lei et al. (2018), which considers that in the non-Newtonian fluid there are two types of particles, namely type 1 and type 2, and the number ratio of particle 1 to 2 is 3:7. In describing the interaction between particles, the parameters ($A_{11}, A_{22}, B_{11}, B_{22}$) in Eq. (2) are for interaction between particles with the same property, while A_{12} and B_{12} are for that between different particles. The values of these parameters in MDPD unit and the corresponding real physical units are shown in Table 1, where A_{ij} refers to the attraction parameter and B_{ij} refers to the repulsive force parameter.

When constructing non-Newtonian droplets by using MDPD, the consistency coefficient and non-Newtonian index in the constitutive equation can be changed by adjusting the dissipative force coefficient (γ) between particles. Here we verify the non-Newtonian property of this fluid by means of reversed Poiseuille flow test. As shown in Fig. 2(a), we fill up the type 1 and type 2 particles under the number ratio of 3:7 in a specific domain with the size of $16 \times 12 \times 12$, then divide the domain into two equal parts along the x -axis, and apply equal and opposite forces a_z on these two parts in z -direction. D_p , whose value is 8, is the half length of the simulation domain along the x -direction. The periodic boundary conditions of this simulation domain are as below: when $x = 0, v_z = 0$; when $x = -8, v_z = 0$; when $x = 8, v_z = 0$.

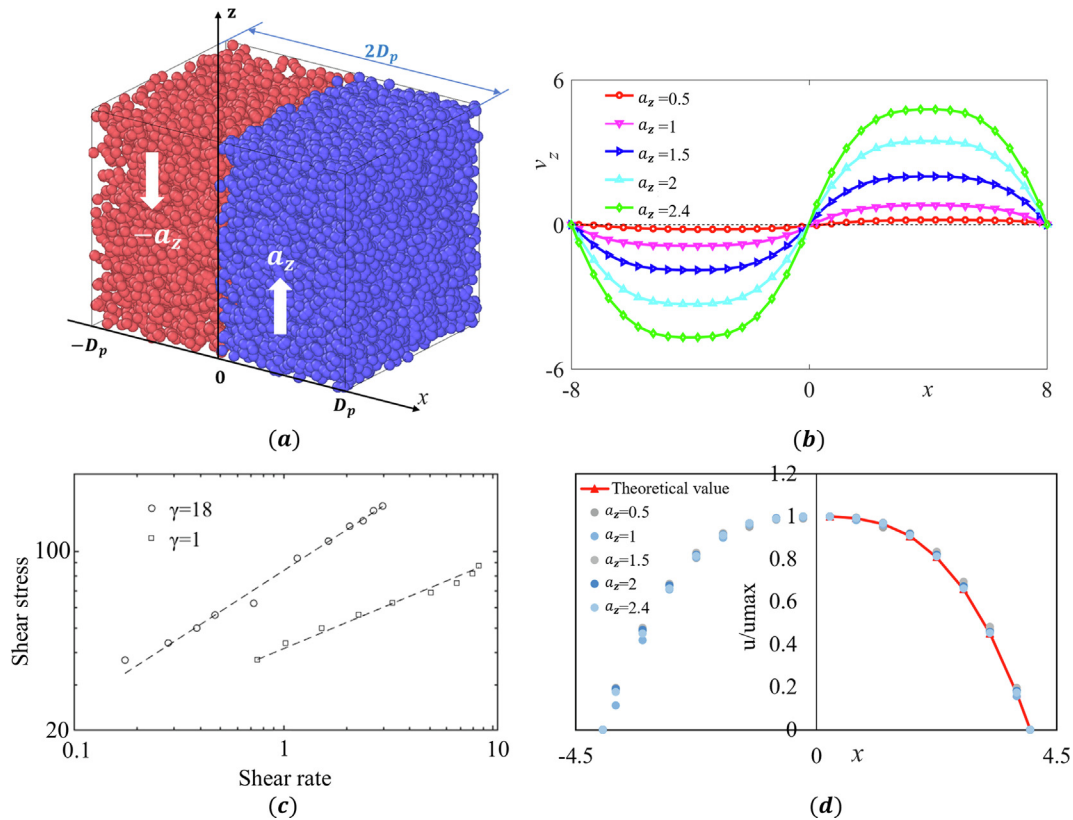
Fig. 2(b) shows the distribution of v_z along the x -direction after applying external forces with different intensities. The parabolic velocity distribution along the simulation domain includes part of the velocity platform flow near the center, which is different from that in the case of Newtonian fluid. And the greater the external force, the smaller the velocity platform. When using the reversed Poiseuille flow to verify the non-Newtonian property of fluid by MDPD, the slope of the $v_z - x$ curve at $x = 0$ can be used to calculate the shear rate $\dot{\gamma}$, with the shear stress $\tau_s = \rho a_z D_p$, where ρ, a_z and D_p are respectively the non-Newtonian fluid density, the acceleration of a fluid of unit mass produced by an external force along the z -direction, and half length of the simulation domain along the x -direction. The relationship between shear rate and shear stress can be found in Fig. 2(c), where the relationship between shear stress and shear rate conforms to the power law ($\tau_s = K\dot{\gamma}^n$), $\tau_s = 83.48\dot{\gamma}^{0.53}$ ($\gamma = 18$) and $\tau_s = 41.39\dot{\gamma}^{0.34}$ ($\gamma = 1$). In these two correlations, the power law exponent n is < 1 , which also proves the pseudoplastic of the fluid.

Fig. 2(d) shows the comparison between the simulated and theoretical values of u/u_{max} in the reversed periodic Poiseuille flow. The scatters in the plot mark the simulated values of u/u_{max} obtained by the reversed Poiseuille flow (Fig. 2(b)), and the red line represents the analytical solution of u/u_{max} , whose formula $u/u_{max} = 1 - 0.0183x^{2.89}$ is derived from $\tau_s = 83.48\dot{\gamma}^{0.53}$ ($\gamma = 18$). The result shows that the simulated values are close to the theoretical values, with a maximum relative error around 7 %, which appears near the wall, indicating that the simulation method describing the non-Newtonian property in this work is valid.

Table 1

Particle interaction parameters of non-Newtonian droplet in MDPD unit and the corresponding real physical units.

	A_{11}	A_{22}	A_{12}	B_{11}	B_{22}	B_{12}	r_c	r_d
MDPD units	-50	-50	-60	25	25	12	1.25	0.9
Physical units ($\times 10^{-7}$)	-4.97 N	-4.97 N	-5.97 N	2.49 N	2.49 N	1.19 N	16.5 m	11.9 m

**Fig. 2.** (a) Schematic diagram of the reversed periodic Poiseuille flow method; (b) Velocity profile of non-Newtonian fluid ($\gamma = 18$) in periodic Poiseuille flow. Black dash line present $v_z = 0$; (c) Rheological curve of the fluid. (d) Comparison between the simulated and theoretical values of u/u_{max} in reversed periodic Poiseuille flow.**Table 2a**

Parameters of Non-Newtonian droplet in MDPD unit and the corresponding real physical units.

Description	MDPD units	Physical units
Diameter, D_0	17	2.25×10^{-5} m
Density, ρ	7.79	1350 kg/m ³
Surface tension, σ	46.54	0.35 N/m
Dynamic viscosity, μ	$\mu_a = 41.39\dot{\gamma}^{-0.66}$ $\mu_a = 83.48\dot{\gamma}^{-0.47}$	$\mu_a = 0.054\dot{\gamma}^{-0.66}$ Pa s $\mu_a = 0.11\dot{\gamma}^{-0.47}$ Pa s
Weber number, We	0–1 1–100	0.01, 0.03, 0.11, 0.26, 0.46, 0.71 1.02, 1.39, 1.82, 2.30, 2.85, 3.44, 4.10, 4.81, 5.58, 6.4, 11.38, 25.61, 45.53, 71.14, 86.08
Static contact angle, θ_c	100–300 120°	102, 120, 139, 160, 182, 206, 230, 257, 268, 279

2.3. Physical settings

The physical and operational conditions of droplets in this work are shown in Tables 2a and 2b, where the falling droplet and the sessile droplet are of equal size and have the same physical properties.

Since the droplet has inertial force at the beginning of impingement, for convenience the Weber number (We) is introduced to

Table 2b

Parameters of Newtonian droplet in MDPD unit and the corresponding real physical units.

Description	MDPD units	Physical units
Diameter, D_0	17	2.25×10^{-5} m
Density, ρ	6.09	1056 kg/m ³
Surface tension, σ	7.51	0.06 N/m
Dynamic viscosity, μ	7.649	0.0032 Pa s
Weber number, We	6.4, 139	
Static contact angle, θ_c	120°	

characterize the ratio of the droplet inertial force to the surface tension. The specific expression of the droplet Weber number is as follows:

$$We = \frac{\rho U_0^2 D_0}{\sigma}$$

where ρ is the density of the droplet, U_0 is the initial velocity of the impinging droplet, D_0 is the diameter of the impinging droplet, and σ is the surface tension of the droplet.

This work focuses on the dynamic of a non-Newtonian droplet impinging on a sessile droplet with different Weber numbers, and compares that with two Newtonian droplets. Tables 2a and 2b show the Parameters of Non-Newtonian fluid droplet and Newtonian fluid droplet in MDPD unit and the corresponding real physical units. The Weber number is varied from 6.4 to 182 by changing the initial velocity of impinging droplet.

The relationship between the MDPD units of length (L_{MDPD}), mass (M_{MDPD}) and time (T_{MDPD}) can be mapped to the specific properties of real liquids (i.e. density ρ , surface tension σ and viscosity ν) according to Eq. (7) (Arienti et al., 2011).

$$\begin{aligned} M_{MDPD} &= L_{MDPD}^3 \frac{\rho^*}{\rho} \\ T_{MDPD} &= (M_{MDPD} \frac{\sigma}{\rho})^{0.5} \\ \frac{L_{MDPD}^2}{T_{MDPD}} &= \frac{\nu^*}{\nu} \end{aligned} \quad (7)$$

As an example, the ethylene glycol solution ($\rho^*=1056 \text{ kg/m}^3$, $\sigma^*=56.5 \text{ mN/m}$ and $\nu^*=3.03 \times 10^{-6} \text{ m}^2/\text{s}$) (Son and Kim, 2009) can be matched by $\rho = 6.09$, $\sigma = 7.51$ and $\nu = 0.4$, respectively. So we determine the scaling coefficients between the physical and MDPD units as $L_{MDPD} = 1.32 \times 10^{-6} \text{ m}$, $M_{MDPD} = 4.01 \times 10^{-16} \text{ kg}$, and $T_{MDPD} = 2.31 \times 10^{-7} \text{ s}$ by Eq. (7). Therefore, the actual physical size of the droplet in this work is $2.25 \times 10^{-5} \text{ m}$.

In Tables 2a and 2b, ρ , σ , μ and θ_c are the density, surface tension, dynamic viscosity and static contact angle of the droplet, respectively. The static contact angle is just to represent the hydrophobic property of the solid surface and the contact angle does not participate in the whole simulation calculation. By constructing a non-Newtonian droplet with the number ratio type 1: type 2 = 3:7, the pseudoplastic droplet density ρ is 7.79, and the surface tension σ is 46.54 calculated by the pressure tensor method (Vázquez et al., 2008), using the following formula,

$$\sigma = \int_{-\infty}^{\infty} dz \left[P_{zz} - \frac{1}{2} (P_{xx} + P_{yy}) \right] \quad (8)$$

where P_{xx} , P_{yy} and P_{zz} are the tensors of the liquid film along the x, y and z directions of the system, and dz is the unit length of the liquid film along the z direction.

In this study, the constitutive equations of the non-Newtonian fluid are $\tau_s = 83.48\dot{\gamma}^{0.53}$ ($\gamma = 18$) and $\tau_s = 41.39\dot{\gamma}^{0.34}$ ($\gamma = 1$). Therefore, the corresponding relationship between the apparent viscosity of the droplet and the shear rate are as follows: $\mu_a = 83.48\dot{\gamma}^{-0.47}$ ($\gamma = 18$) and $\mu_a = 41.39\dot{\gamma}^{-0.66}$ ($\gamma = 1$).

In this work, for the droplet with corresponding density, size and surface tension ($\rho = 1350 \text{ kg/m}^3$, $D_0 = 2.25 \times 10^{-5} \text{ m}$, $\sigma = 0.35 \text{ N/m}$), the Bond number, $Bo = \frac{\rho g D_0^2}{4\sigma}$, is about 4.78×10^{-6} (g is the gravitational acceleration), which means that the ratio of liquid gravity and surface tension is in order of 10^{-6} . And the Galileo number, $Ga = \frac{D_0^3 \rho^2 g}{\mu^2}$, which characterize the ratio between gravitational acceleration and viscous force ($\mu = 9.06$ under $\dot{\gamma} = 10$), is as small as 1.4×10^{-3} . Therefore, in this work we ignored the effect of gravity force on droplet.

3. Results and discussion

3.1. Dynamic process of a single non-Newtonian droplet impinging a sessile droplet

Fig. 3 shows impingement process between two non-Newtonian droplets with different Weber numbers, where the very right comparison snapshot is the equilibrium state of a merged droplet from two Newtonian droplets whose properties are shown in Tables 2a and 2b. Fig. 3(a) and (c) show the front view of the droplets, while the Fig. 3(b) and (d) show the cross section view of the droplets by intercepting a thin particle layer along the y-axis ($y \in [-0.5, 0.5]$). During the impingement ($We = 6.4$, Fig. 3(a)), the liquid at the interface of the two droplets is squeezed and bulges at the edge ($t_{mdpd} = 10$). However, the inertial force of the impinging droplet is not large enough to result in a continuous spreading of the sessile droplet, and the bulge subsides quickly. As a result, the shape of the sessile droplet does not change significantly compared with its initial state. In the case of $We = 139$ (Fig. 3(c)), the falling droplet has a greater inertial force to overcome the viscous and interfacial forces, so that the bulge at the droplet interface will not retract and the merged droplet evolves into a flat shape ($t_{mdpd} = 10$). Afterwards, the droplet continues to retract ($t_{mdpd} > 10$) under the influence of the hydrophobic surface, and finally reaches stability ($t_{mdpd} = 50000$). From the comparison between these two cases, it can be found that the stable shape of non-Newtonian droplets varies with the Weber number in the impingement, which is different from the result of Newtonian droplets where the stable shape of the Newtonian droplet generally does not change with the Weber number of the impinging droplet.

In order to quantitatively describe the influence of impinging droplet We on the stable shape of the merged droplet, a normalized variable H/L is used,

$$\frac{H}{L} = \frac{H_{eq}/L_{eq}}{H_{s0}/L_{s0}}$$

where H_{eq} and L_{eq} are respectively the height and width of the merged droplet when it is stable, and H_{s0} and L_{s0} are respectively the height and width of the sessile droplet before the impingement, as shown in Fig. 4. Obviously, $H/L < 1$ indicates that the droplet becomes flat, and $H/L > 1$ indicates a tall droplet.

Fig. 5 shows H/L as a function of Weber number of impinging droplet. It can be seen that for the non-Newtonian droplets, both the Weber number and the apparent viscosity μ_a of the droplet can affect value of H/L . Scatters marked by red squares and blue circles represent the non-Newtonian fluids with different viscosities. By increasing the Weber number or decreasing the apparent viscosity μ_a , the merged droplet achieves low scaled H/L . The solid line represents the fitting line of the corresponding fluid. The dashed line of the same color represents the H/L of the stationary large droplet on the surface, whose volume is the sum of those of the falling droplet and the sessile droplet. The hollow circle is the intersection of the fitting line and the dashed line, and the corresponding Weber number at the intersection is defined as the critical Weber number We_c , which indicates the critical condition to overcome the resistance of droplet deformation during the impingement.

3.2. Mixing process of two non-Newtonian droplets

3.2.1. Mixing process

In observing the mixing process of two non-Newtonian droplets, we intercept a thin particle layer along the y-axis ($y \in [-0.5, 0.5]$), as shown in Fig. 3(b) and (d). By comparing Fig. 3(a), (c) and Fig. 3(b), (d), which shows the particle moving on the dro-

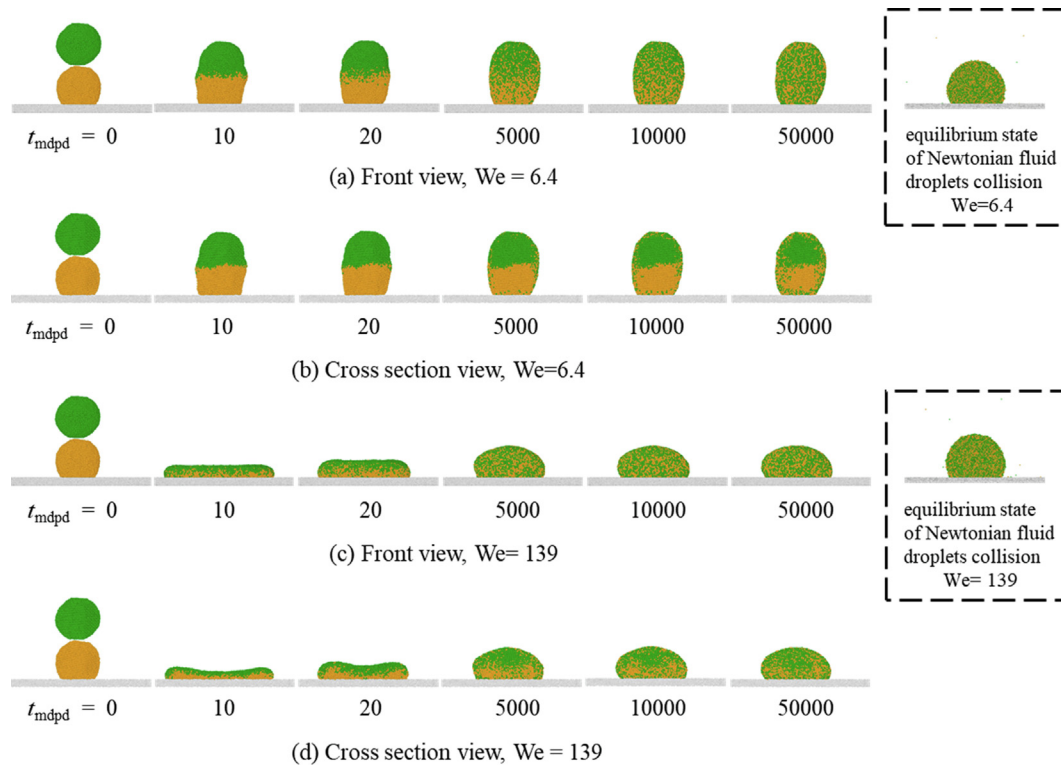


Fig. 3. Snapshots of the impingement of a non-Newtonian droplet and a sessile droplet with different Weber numbers (a) $We = 6.4$ (front view), (b) $We = 6.4$ (cross section view), (c) $We = 139$ (front view), and (d) $We = 139$ (cross section view) and the last column is the equilibrium state of Newtonian fluid droplets after impingement.

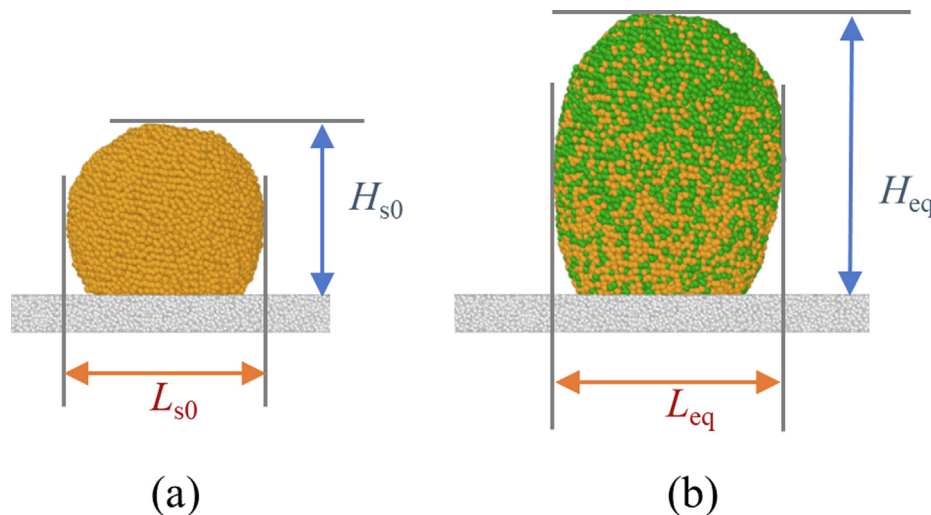


Fig. 4. (a) The initial state of the sessile droplet and (b) the equilibrium state of the merged droplets.

plet surface, it can be seen that although on the surface of the droplet it has achieved a relatively uniform mixing (Fig. 3(a), $We = 6.4$, $t_{mdp} = 50000$; Fig. 3(c), $We = 139$, $t_{mdp} = 5000$), inside the droplet there is still a large area without fully mixed (Fig. 3(b), $We = 6.4$, $t_{mdp} = 50000$; Fig. 3(d), $We = 139$, $t_{mdp} = 5000$).

In Fig. 3(b), for the case $We = 6.4$, the interface between the two droplets after the impingement is still clear ($t_{mdp} = 10 - 20$), indicating that the particles inside the droplet are less affected by the inertial force, and there is no significant mutual penetration between the particles of the two droplets. As the mixing goes on, the particles of the impinging droplet will first spread on the surface of the sessile droplet, and then diffuse from the surface of

the sessile droplet to the interior ($t_{mdp} = 20 - 50000$). In general, there are some differences of the mixing process after the head-on collision between two Newtonian droplets and two non-Newtonian droplets, even if the durations of convection dominant stage are both much shorter than that of diffusion dominant stage: for the Newtonian droplets, the particle diffusion starts at the impinging interface of the two droplets and then goes on along the direction of particle concentration gradient (Yi et al., 2020), whereas in the mixing of two non-Newtonian droplets, the particles of the impinging droplet will first spread on the surface of the sessile droplet, and then diffuse from the surface of the sessile droplet to the interior.

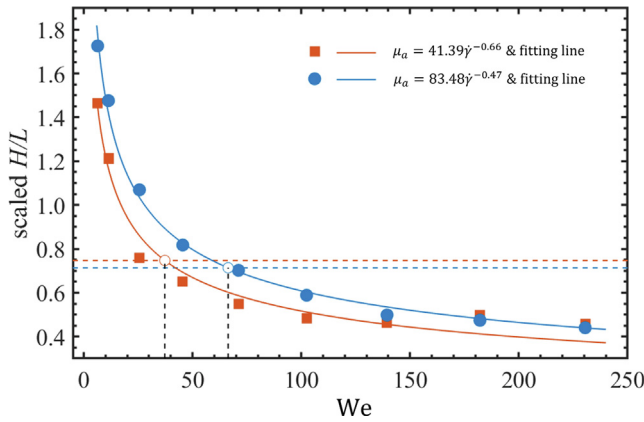


Fig. 5. The H/L of merged droplet versus We .

In Fig. 3(d), for the case $We = 139$, the impinging droplet has a higher initial velocity, that is, the inertial force is larger. As a result, the deformation of the droplet will be more intensive, which will lead to the irregular arrangement of the particles, so that the particles at the interface of the two droplets begin to mix with each other ($t_{\text{mdpd}} = 10 - 20$). We can see from the Fig. 3(b) and (d), when t_{mdpd} is 50000, the particles from two droplets mix well, while the droplet under $We = 6.4$ is still far from this state.

In order to quantitatively describe the mixing process of droplets, we divide the mixing process into the convection-dominated stage and diffusion-dominated stage according to the particles' velocity and kinetic energy of the impinging droplet, and these two stages are discussed respectively in sections 3.2.2 and 3.2.3.

3.2.2. Particle behavior in the convection-dominated mixing stage

In the judgement and determination of the duration of convection-dominated mixing stage, the kinetic energy of the droplets is a useful indicator. Without considering gravity, the energy of two-droplet system should satisfy Eq. (9):

$$E_k^0 + E_s^0 + A_{sv}\sigma_{sv} = A_{lv}\sigma_{lv} + A_{ls}\sigma_{ls} + (A_{sv} - A_{ls})\sigma_{sv} + E_k + E_v \quad (9)$$

In Eq. (9), the left side is the total energy of the system at the initial moment, and the right side is the total energy of the system at any moment t_{mdpd} . E_k^0 and E_s^0 are respectively the initial kinetic energy and initial surface energy of the system. A_{lv} , A_{ls} and A_{sv} respectively represent the gas-liquid interface, solid-liquid interface and initial gas-solid interface area in the system. Correspondingly, σ_{lv} , σ_{ls} and σ_{sv} are respectively the interfacial tension at the gas-liquid interface, the solid-liquid interface, and the gas-solid interface. E_k and E_v are the kinetic energy and dissipated energy at moment t_{mdpd} , respectively.

Obviously, since the surface energy of solid surface $A_{sv}\sigma_{sv}$ before and after the impingement is the same, then

$$E_k^0 + E_s^0 = E_k + E_s + E_v \quad (10)$$

where the droplet surface energy E_s is

$$E_s = A_{lv}\sigma_{lv} + A_{ls}(\sigma_{ls} - \sigma_{sv}) \quad (11)$$

According to the Young's equation ($\sigma_{ls} - \sigma_{sv} = -\sigma_{lv}\cos\theta_c$), there is

$$E_s = A_{lv}\sigma_{lv} - A_{ls}\sigma_{lv}\cos\theta_c \quad (12)$$

where θ_c is the static contact angle.

The droplet kinetic energy E_k can be obtained by Eq. (13):

$$v = \frac{1}{N} \sum_{i=1}^N v_i$$

$$E_k = \frac{1}{2}M|v_{\text{mean}}|^2 = \frac{1}{2}M(v_{x,\text{mean}}^2 + v_{y,\text{mean}}^2 + v_{z,\text{mean}}^2) \quad (13)$$

where v_x , v_y , v_z represent the velocity components of a single particle in the droplet along the x, y, and z directions. $v_{x,\text{mean}}$, $v_{y,\text{mean}}$ and $v_{z,\text{mean}}$ are respectively the average value of the particles velocities in these three directions. M is the mass of droplet. Since the mass of each particle in the droplet is set to 1 in the MDPD simulation, the droplet mass is numerically equal to the total number of particles in the droplet.

For convenience, the normalized kinetic energy of the impinging droplet, E_k/E_k^1 , is calculated, where E_k is the actual kinetic energy of the impinging droplet, and E_k^1 is the kinetic energy when the droplet has a unit velocity. In this work, it is considered that when the kinetic energy of the impinging droplet reaches as low as $E_k/E_k^1 < 10^{-3}$, the corresponding duration from the beginning of impingement is the that of convection-dominated mixing stage (t_c). And after that ($t_{\text{mdpd}} > t_c$), the molecular diffusion-dominated mixing stage starts.

Fig. 6 shows the particle distribution on the droplet surface (front view) and inside the droplet (cross section view) at $t_{\text{mdpd}} = t_c$ when the Weber number of impinging droplet is 6.4 and 139. It can be seen that at the end of the convection-dominated mixing stage, for the case $We = 6.4$, the interface between the two droplets is still clear, while for the Weber number 139, there is plenty of particles from impinging droplet both on the sessile droplet surface and inside it, indicating a more significant mass transfer in the impinging and mixing process.

It can be seen from the slope of velocity-time curve (Fig. 7(a)) that the slope of this curve for the impinging droplet under $We = 139$ is larger, compared with that in the case $We = 6.4$, which indicates that the larger the Weber number of the impinging droplet, the faster its velocity decreases. The same phenomena can also be found in the change of kinetic energy of the impinging droplet.

Fig. 8 shows the change of convection dominated stage duration t_c with the increasing Weber number, with three repetitions by changing random seeds under same conditions. As can be seen Fig. 8, the relationship between t_c and Weber number can be mainly divided into three sections, which reflects the competitive relationship between inertial force and surface tension. In the region $We \in [0.05, 1]$, the kinetic energy of impinging droplet is too small to overcome the surface energy of the sessile droplet, and therefore the convection is weak, resulting in the fluctuation of t_c . In the region $We \in (1, 205]$, the kinetic energy can generally overcome the surface energy as the increase of Weber number, and the convection becomes significant. Meanwhile, the competition between droplet inertial force and surface tension tends to keep stable in this section. In the region $We \in (205, 280]$, the large kinetic energy can be destructive to the integrity of the droplet interface. As shown in Fig. 9, when the Weber number reaches 280, the droplet eventually fails to keep unbroken after the impingement, and therefore the duration of convection dominated stage can be unpredictable or even infinite due to its breakup into tens of tiny daughter droplets as shown in Fig. 10.

In order to realize the concentration field in the droplet, we intercept a thin particle layer along the y-axis ($y \in [-0.5, 0.5]$), as shown in Fig. 11. It can be seen that the concentration field of three phases have different changes with time growth. In the region $We \in [0.05, 1]$, there is no significant mixing between two droplets, while in the region $We \in (1, 205]$, there presents a mixed layers with concentration gradient along the impinging direction; in the region $We \in (205, 280]$, there is rapid mixing as soon as the happening of impingement, and there is finally a relatively uniform mixing.

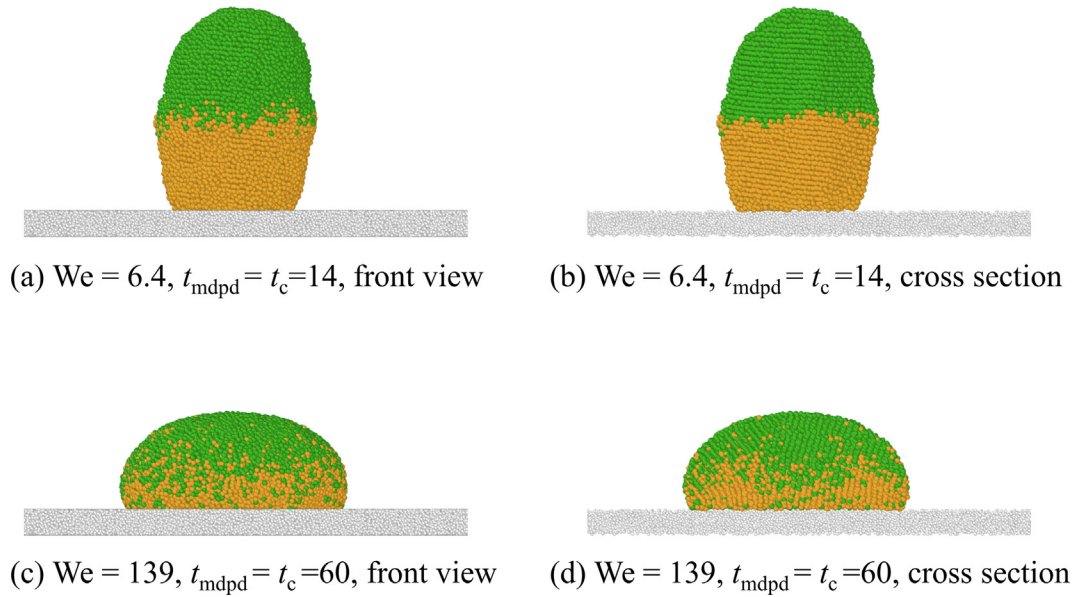


Fig. 6. The snapshots of the droplets in (a) front view and (b) the cross section view under $We = 6.4$ at the moment when $t_{\text{mdpd}} = t_c$; the snapshots of the droplets in (c) front view and (d) the cross section view under $We = 139$ at the moment when $t_{\text{mdpd}} = t_c$.

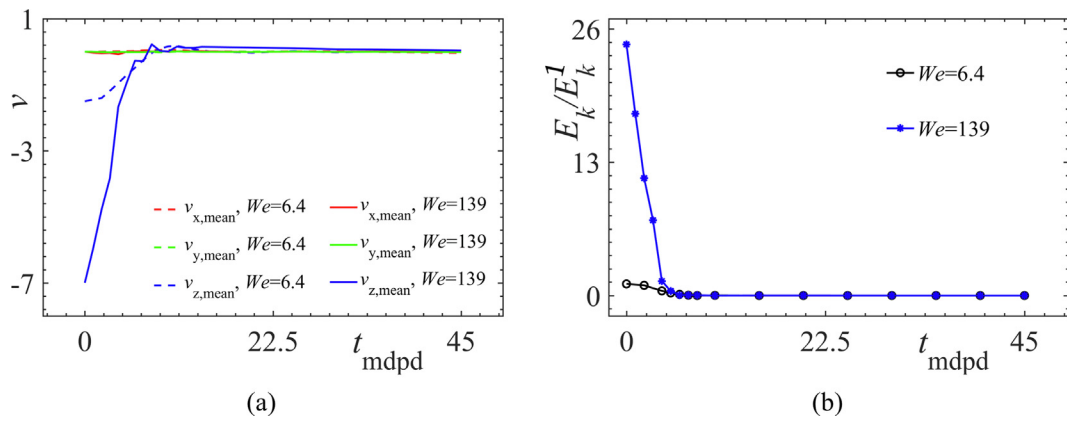


Fig. 7. (a) The velocity $v_{x,\text{mean}}$, $v_{y,\text{mean}}$, $v_{z,\text{mean}}$ and (b) the scaled kinetic energy E_k/E_k^1 of the impinging non-Newtonian droplet as a function of time.

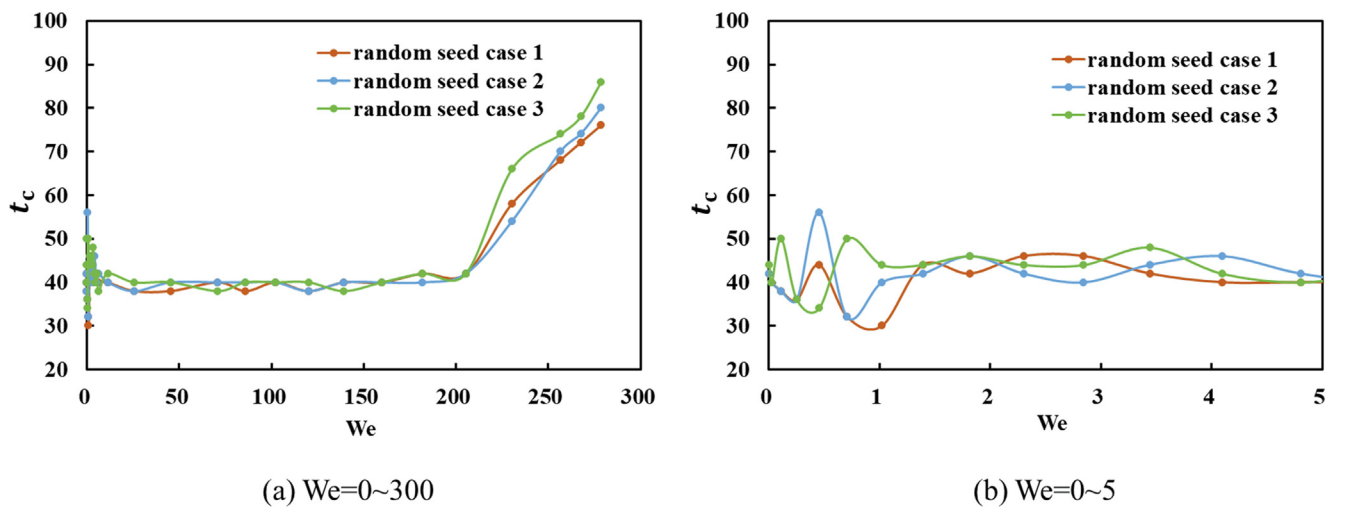


Fig. 8. (a) The change of t_c with the increasing Weber number and (b) the enlarged view of (a) ($We \in [0.05, 1]$).

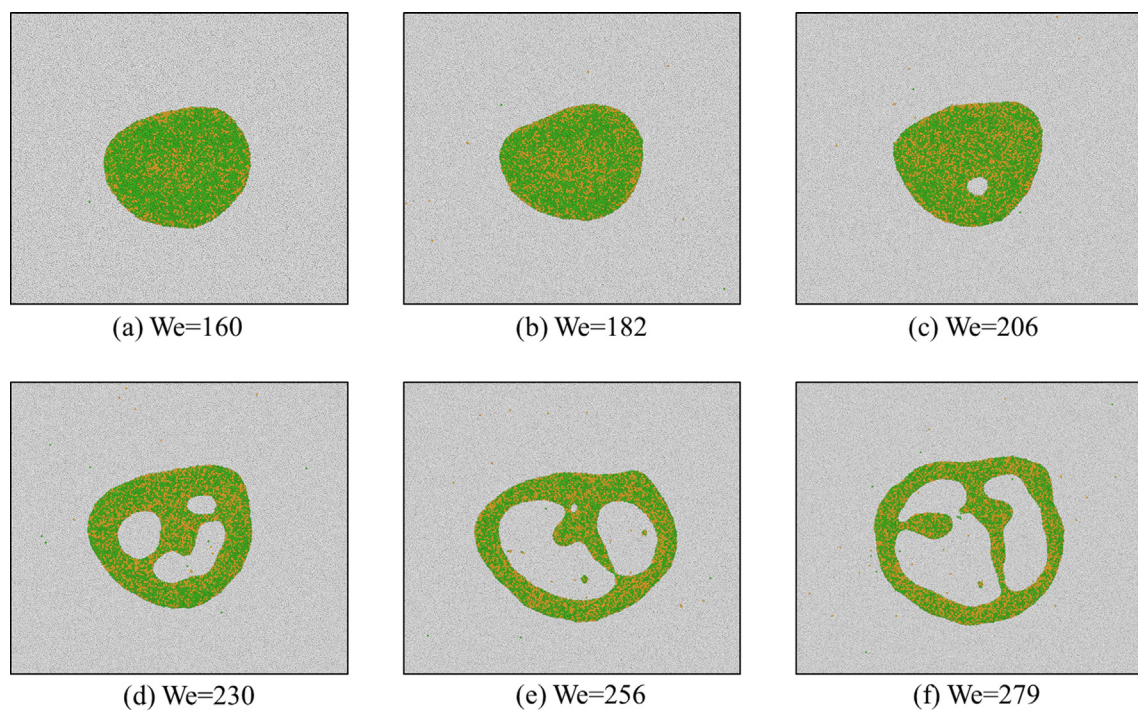


Fig. 9. Top view of the mixed droplet under different impinging droplet Weber numbers ($t_{\text{mdd}} = 18$) with (a) $We = 160$, (b) $We = 182$, (c) $We = 206$, (d) $We = 230$, (e) $We = 256$ and (f) $We = 279$.

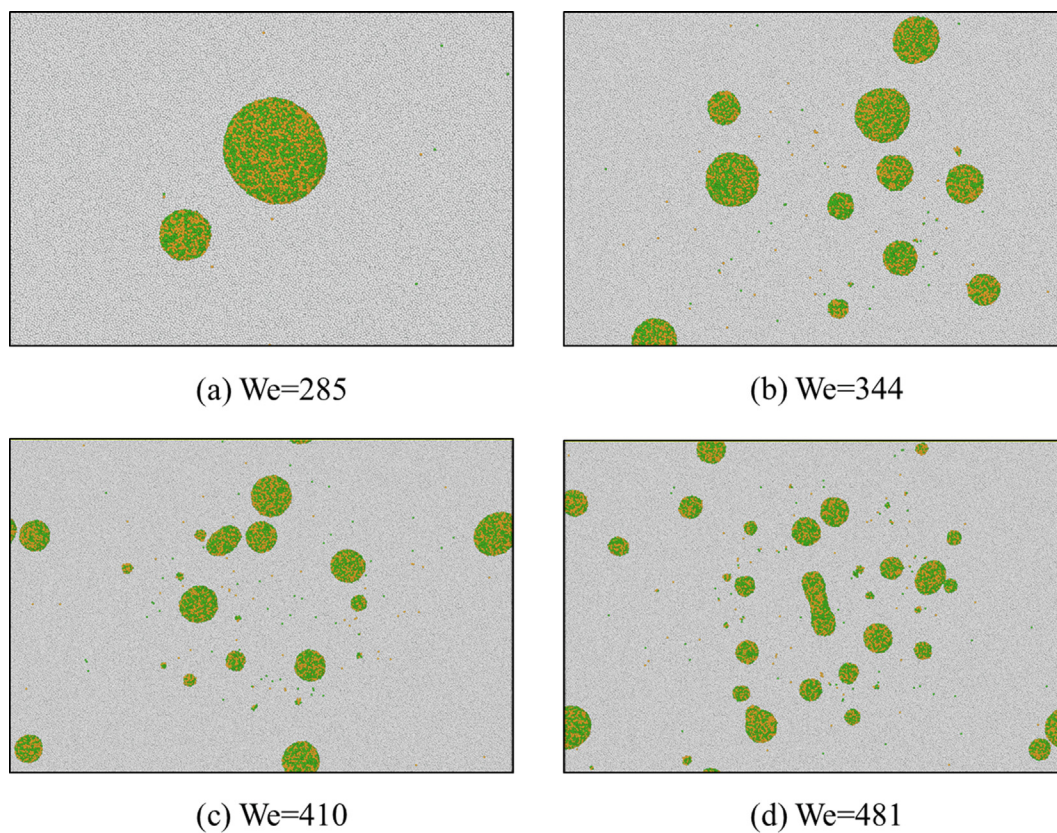


Fig. 10. The breakup of droplet after impingement and resulting tiny daughter droplets ($t_{\text{mdd}} = 96$) ($We \in [285, 481]$).

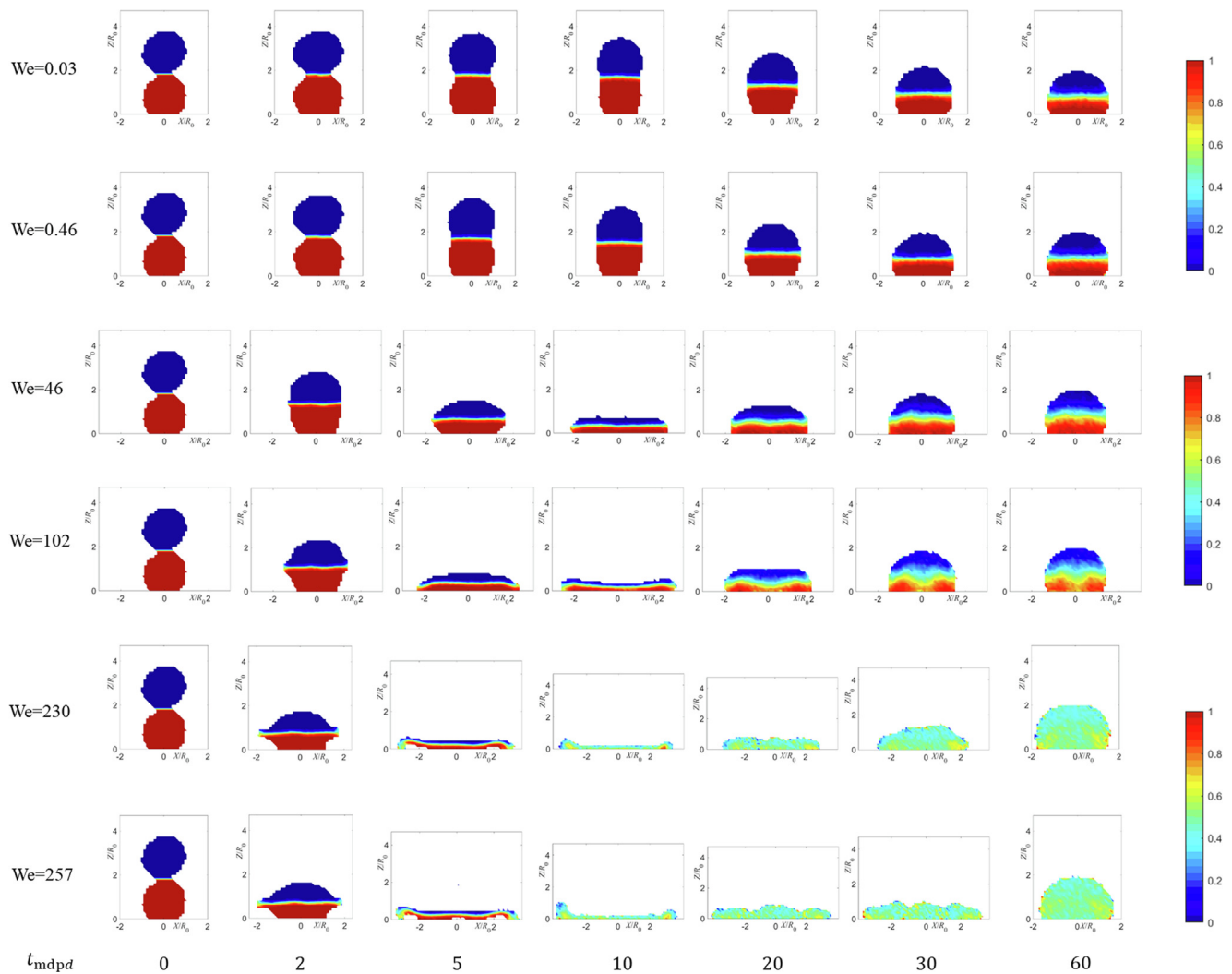


Fig. 11. The change of concentration field during the mixing of two droplets under various Weber numbers ($t_{\text{mdpd}} \in [0, 60]$).

To sum up, when the Weber number is lower than 1, the mixing of the two non-Newtonian droplets is so poor, mainly dependent on the molecular diffusion. When the Weber number is greater than 205, t_c increases with the increase of the Weber number, which is not beneficial to the rapid mixing. In practice, in order to reach a controllable mixing, we should make sure that the Weber number is between 1 and 205.

In order to study the effect of Weber number on the kinetic energy dissipation rate of impinging droplets, we fixed the appar-

ent viscosity μ_a of the impinging droplet, $\mu_a = 41.39\dot{\gamma}^{-0.66}$, and changed the Weber number of impinging droplet from 6.4 to 182. Then the average energy dissipation rate $\frac{E_k}{E_k t_c}$ of the impinging droplet, denoted by $\Delta E_k^*/t_c$, is shown in Fig. 12. This average kinetic energy dissipation rate is calculated from the change of droplet kinetic energy over the duration of convection dominated mixing stage. The relationship between kinetic energy dissipation rate and the Weber number of impinging droplet is regressed as Eq. (14).

$$\frac{\Delta E_k^*}{t_c} = 0.266 \ln(We) - 0.2787 (We \in [6.4, 182]) \quad (14)$$

Weber number and ΔE_k^* ($\Delta E_k^* = v_z^2$) are related with the velocity of the impinging droplet (v_z), so that t_c could be calculated from Eq. (14). Therefore, in practice, we can make t_c under control by setting an appropriate velocity of the impinging droplet.

3.2.3. Particle behavior in the molecular diffusion-dominated mixing stage

As mentioned in section 3.2.1, when the mixing is in the molecular diffusion-dominated mixing stage ($t_{\text{mdpd}} > t_c$), the particles in the impinging droplet will first diffuse from the surface of the impinging droplet to the surface of the sessile droplet, and then diffuse from the surface of the sessile droplet to the interior. Here

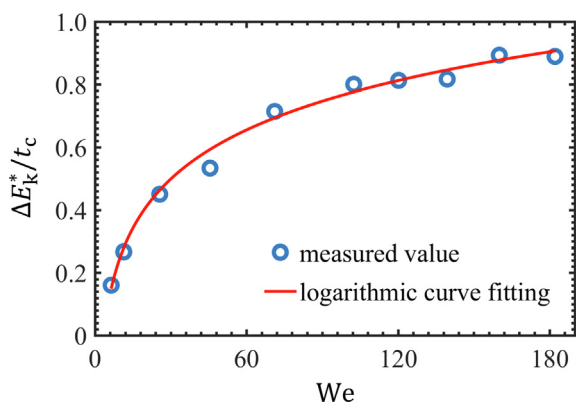


Fig. 12. The relationship between the $\Delta E_k^*/t_c$ of the impinging droplet and We .

we take the case Weber number of impinging droplet 6.4 as an example to analysis this process.

In section 2.2, we have introduced that in the construction of a non-Newtonian droplet there are two types of particles (type 1 and type 2), and the number ratio of particle type 1 to type 2 is 3:7. In order to differentiate these two types particle in the impinging and

sessile droplets, type 1 particles in the impinging and sessile droplets are labeled Group 1 and Group 3, respectively, and the type 2 particles are labeled Group 2 and Group 4, respectively. These four groups are marked by 4 colors, as shown in Fig. 13.

It can be seen from Fig. 13 that the type 1 particles move slowly after the formation of interface in the impingement. When

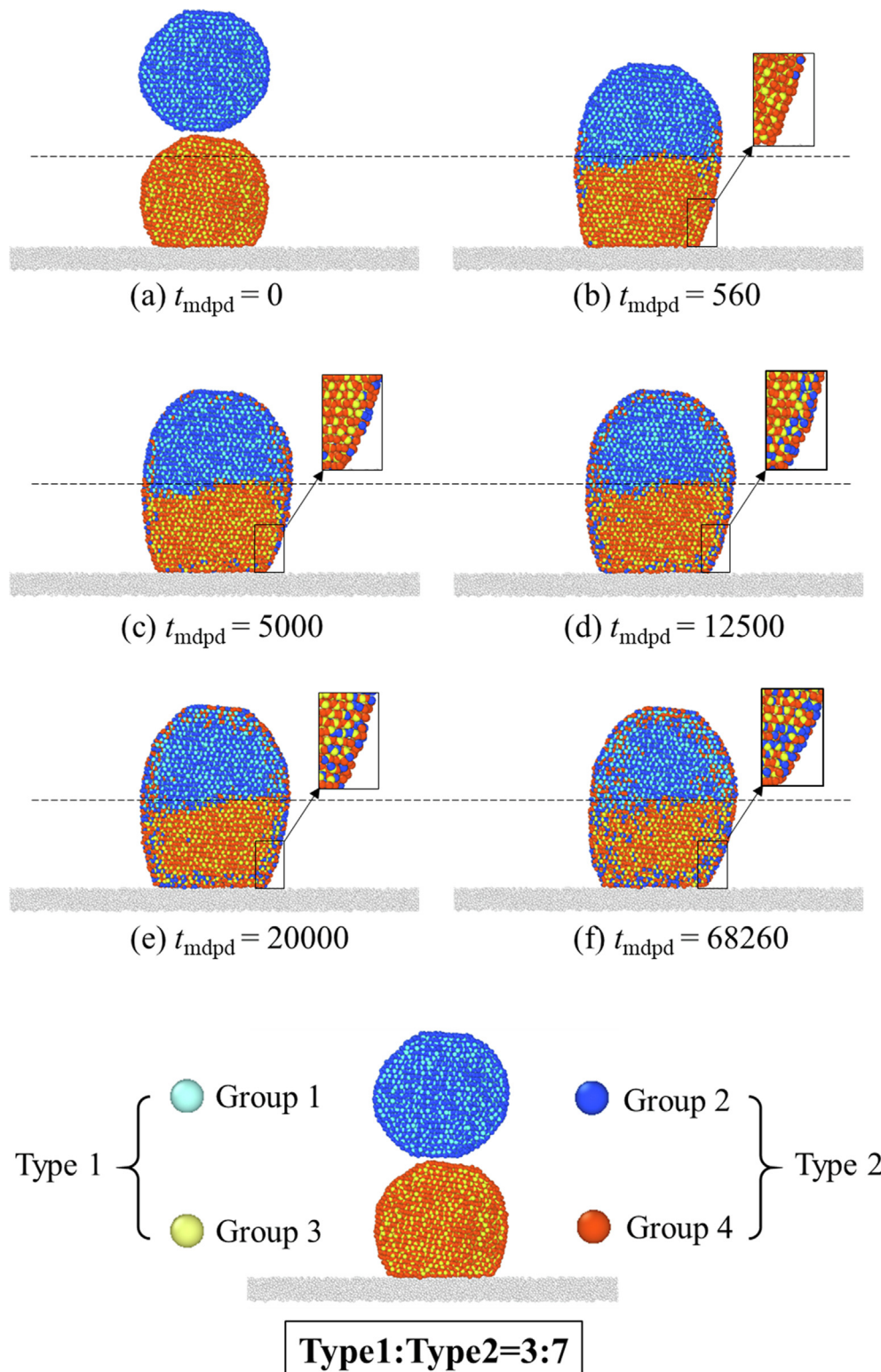


Fig. 13. Cross section views of four groups of particles in the diffusion-dominated stage ($We = 6.4$). The black dash line between droplets represents the interface just after impingement.

$t_{\text{mdpd}} < 2000$, the interface between the two droplets is still clear, and the type 1 particles basically do not penetrate into the sessile droplet at the interface. Molecular diffusion mainly accomplishes with particles type 2: particles type 2 preferentially transports on the sessile droplet surface (from $t_{\text{mdpd}} = 0$ to $t_{\text{mdpd}} = 12500$, Fig. 13 (a-d)), and then particles type 2 on the droplet surface diffuse into the sessile droplet (from $t_{\text{mdpd}} = 12500$ to $t_{\text{mdpd}} = 68260$, Fig. 13 (d-f)). In constructing the non-Newtonian fluid by using two types of particles with a certain number ratio, it is easy for reader to imagine or realize that these two types of particles can be referred to the solute and solvent respectively in the non-Newtonian fluid, which is also described in the literature of Lei et. al (2018). In our understanding, the displacement of any type particles and the redistribution of particles induced by it is due to the energy input by the impinging droplet. As a result, this energy is used to overcome the interaction between particles, which is partly similar to the bead-spring model in describing the behavior of polymers (where the particles type 1 and type 2 behave the beads and the interaction force represents the spring in a certain extent). Therefore, the motion of particles 2 towards the surface of the droplet is the result of energy dissipation to overcome the attractive force between particles.

To explain the diffusion behavior of type 2 particles described above, the diffusion coefficients D of particles type 2 in shear-thinning fluids, which composed of type 1 and type 2 particles, and in fluid with only type 2 particles can be calculated respectively by the Einstein equation (Essajai et al., 2019),

$$D = \frac{1}{6} \lim_{t \rightarrow 0} \frac{d}{dt} (\text{MSD}(t)) \quad (15)$$

where the Mean Squared Displacement (MSD) of freely diffusing particles over time can be expressed as

$$\text{MSD}(t) = \langle |r_i(t) - r_i(0)|^2 \rangle \quad (16)$$

where $r_i(t) - r_i(0)$ is the vector distance traveled by a particle in a certain time interval.

Fig. 14 shows the MSD(t) curves of particles type 2 in two types of fluids mentioned above. In the shear-thinning fluid composed of particles type 1 and type 2, $D_{\text{type 2}/(\text{type 2}+\text{type 1})}$ is 1.67×10^{-4} , while in the fluid with only particles type 2, $D_{\text{type 2}/\text{type 2}}$ is 4.33×10^{-4} . This difference in molecular diffusion coefficient indicates that particles type 2 diffuse faster in the fluid with only particles type 2 than that in the fluid with two components, that is, the diffusion of particles type 2 can be affected by its proportion in the fluid. As a results, in Fig. 13, in the diffusion dominated stage, due to the difference in diffusion coefficient, particles type 2 will preferentially diffuse from the surface of one droplet to that of another one (from $t_{\text{mdpd}} = 0$ to $t_{\text{mdpd}} = 12500$, Fig. 13(a-d)).

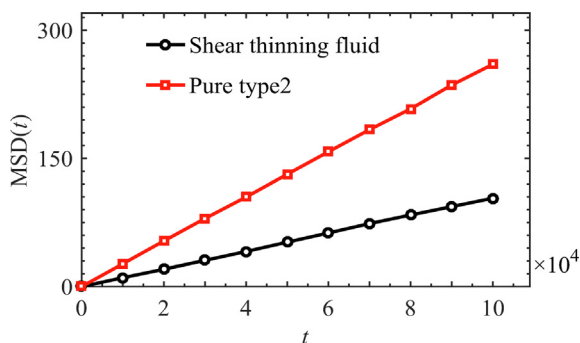


Fig. 14. The variation of MSD(t) of particles type 2 in the shear thinning liquid and pure liquid with only particles type 2.

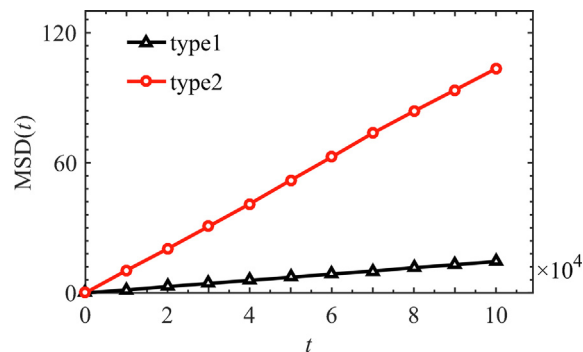


Fig. 15. The MSD(t) curves of particles type 1 and type 2 in the shear thinning fluid.

To explain the phenomenon of particles type 2 on the droplet surface diffusing into the sessile droplet (from $t_{\text{mdpd}} = 12500$ to $t_{\text{mdpd}} = 68260$, Fig. 13(d-f)), the MSD(t) curves for particles type 1 and for particles type 2 in a shear-thinning fluid with both type 1 and type 2 particles are shown in Fig. 15. According to the MSD(t) curves, in the shear-thinning fluid, the diffusion coefficient of particles type 1 $D_{\text{type 1}/(\text{type 1}+\text{type 2})}$ is 1.67×10^{-5} , while the diffusion coefficient of particles type 2 $D_{\text{type 2}/\text{type 2}}$ is 1.67×10^{-4} , which means that the diffusion coefficient of particles type 1 is much lower than that of particles type 2. The amount of particles type 2 is about twice that of type 1 particles, and the attraction between particles of different types is greater than the attraction between particles of the same type. This results in a larger resistance on the diffusion of particles type 1, which slows down their motion. The mixing process of the two droplets is dominated by the diffusion of type 2 particles.

3.3. Mixing time and mixing degree of non-Newtonian droplets

In the determination of mixing time for the non-Newtonian droplets reaching the complete mixing after impingement, the mixing function proposed by Pak et al. (2020), is used to calculate the total mixing time t_{total} . By calculating the average of the square of particle coordinates ($\langle x(t)^2 \rangle$, $\langle y(t)^2 \rangle$ and $\langle z(t)^2 \rangle$) in three dimensions, the particles can be regarded as completely mixed when the $m_x(t) = m_y(t) = m_z(t) = 1$. Since the non-Newtonian droplets are composed of two different types of particles, the following two forms can be used when plotting the curve of the mixing degree function $m(t)$ versus time:

(1) By calculating $m(t)$ of type 1 particles, as in Eq. (17):

$$\begin{aligned} m_{x,\text{type 1}}(t) &= \langle x(t)^2 \rangle_{\text{type 1}} / \langle x(t)^2 \rangle_{\text{all}} \\ m_{y,\text{type 1}}(t) &= \langle y(t)^2 \rangle_{\text{type 1}} / \langle y(t)^2 \rangle_{\text{all}} \\ m_{z,\text{type 1}}(t) &= \langle z(t)^2 \rangle_{\text{type 1}} / \langle z(t)^2 \rangle_{\text{all}} \end{aligned} \quad (17)$$

(2) By calculating $m(t)$ of type 2 particles, as in Eq. (18):

$$\begin{aligned} m_{x,\text{type 2}}(t) &= \langle x(t)^2 \rangle_{\text{type 2}} / \langle x(t)^2 \rangle_{\text{all}} \\ m_{y,\text{type 2}}(t) &= \langle y(t)^2 \rangle_{\text{type 2}} / \langle y(t)^2 \rangle_{\text{all}} \\ m_{z,\text{type 2}}(t) &= \langle z(t)^2 \rangle_{\text{type 2}} / \langle z(t)^2 \rangle_{\text{all}} \end{aligned} \quad (18)$$

Here, 'type 1' and 'type 2' are respectively the parts of the types of particles, and 'all' represents the merged droplet part.

Fig. 16 shows the variation of the mixing function $m(t)$ with time when changing the Weber number of impinging droplet, where Fig. 16(b), (d), (f), (h) are the partial enlarged figures (t_{mdpd}

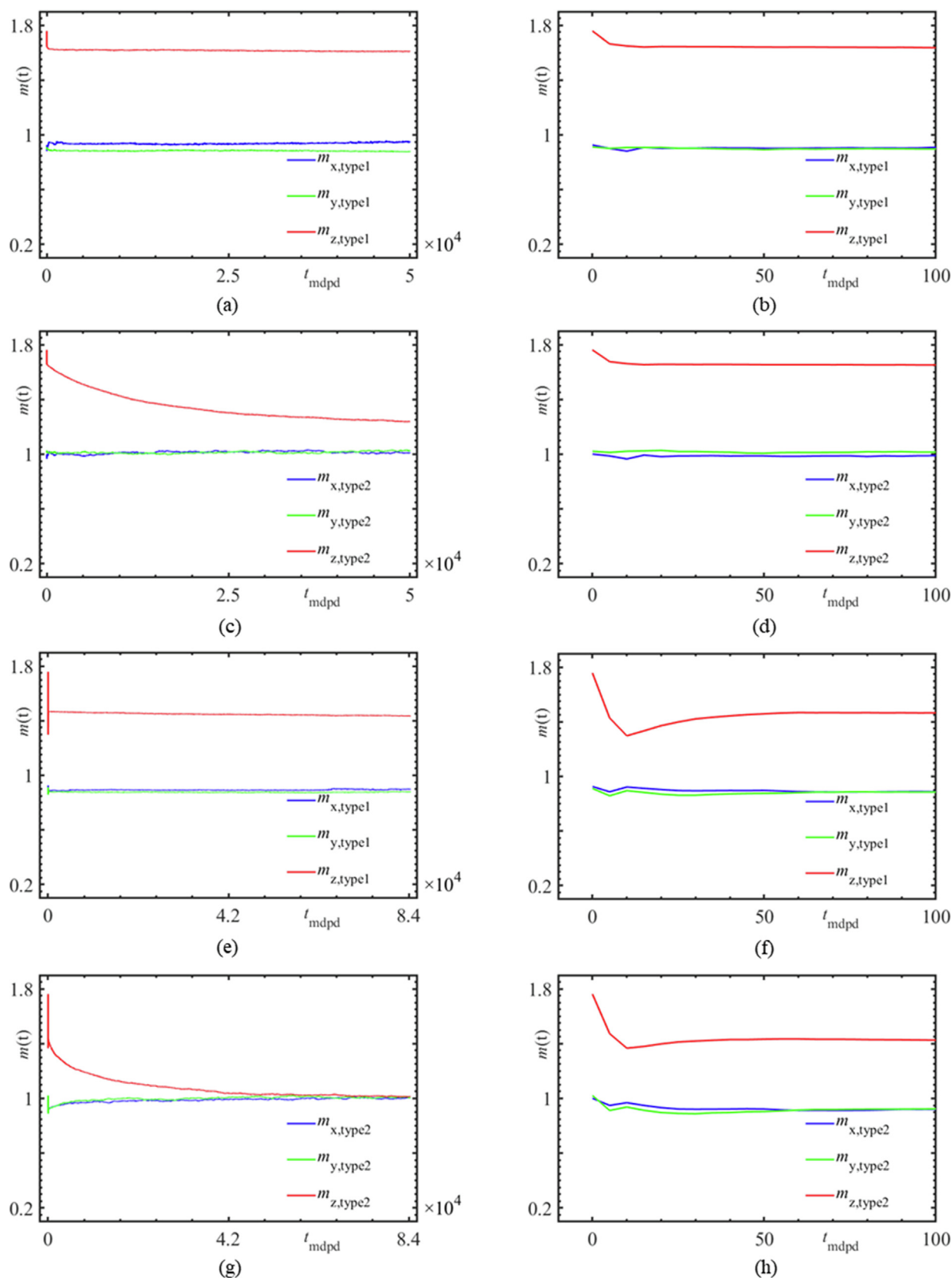


Fig. 16. The change of $m(t)$ along the time with (a-d) $We = 6.4$ and (e-h) $We = 139$. (b), (d), (f) and (h) are the partial enlarged figures (t_{mdpd} from 0 to 100) of (a), (c), (e) and (g) respectively.

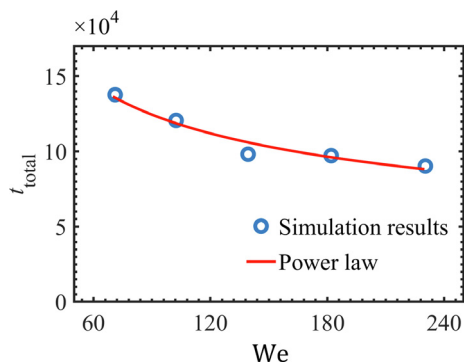


Fig. 17. The relationship between the minimum time required for the droplets to reach complete mixing and We .

from 0 to 100) of Fig. 16(a), (c), (e), (g) respectively. Due to the symmetry of “head-to-head” impingement, the mixing function in x - and y - direction $m_x(t)$ and $m_y(t)$ do not change much after the impingement, fluctuating around $m(t) = 1$. On the contrary, there is significant displacement of particles of the two droplets along the z direction, as a result, $m_z(t)$ changes obviously with time.

When the Weber number of impinging droplet is 6.4, the value of $m_z(t)$ starts to decrease as soon as the impingement happens, but the change of $m_z(t)$ is different for particles: for type 1 particles, $m_{z,type 1}(t)$ decreases slowly with time ($t_{mdpd} \in [1 \times 10^4, 5 \times 10^4]$, $m_{z,type 1}(t)$ value only decreases by 2 %); while the $m_{z,type 2}(t)$ of type 2 particles changed relatively obviously ($t_{mdpd} \in [1 \times 10^4, 5 \times 10^4]$, the value of $m_{z,type 2}(t)$ decreased by 20 %). We have known that type 2 particles diffuse faster than type 1 particles from section 3.2.3, and the difference between $m_{z,type 1}(t)$ and $m_{z,type 2}(t)$ in Fig. 16 further confirms this phenomenon.

When the Weber number of impinging droplet is 139, both $m_{z,type 1}(t)$ and $m_{z,type 2}(t)$ start to decrease significantly just after the impingement, which results from the high inertial force of

the impinging droplet and there following a drastically deformation of both droplets (Fig. 3, $t_{mdpd} = 10$). In the retraction stage of the merged droplet (for example, $t_{mdpd} = 100$), the $m_z(t)$ value increases, which can be reflected by the z -coordinate difference of the two droplet particles. The value of $m_z(t)$ increases for the occurrence of the z -coordinate difference of the two droplet particles as the droplet retracts ($t_{mdpd} = 100$). After the start of diffusion dominated stage, the decrease of $m_{z,type 2}(t)$ value is more obvious than that of $m_{z,type 1}(t)$, and when the value of $m_{z,type 2}(t)$ approaches 1 ($t_{mdpd} = 50000$), the value of $m_{z,type 1}(t)$ is still at a high level (about 1.48), which means that the type 2 particles are better mixed compared with type 1 particles.

Here we can determine the minimum time required for a droplet to reach complete mixing by judging whether the $m(t)$ value of type 2 particles keeps relatively stable around 1.0, $|1 - m_{type 2}(t)| \leq 0.01$, which is due to the random force introduced into the MDPD.

Fig. 17 shows the change of t_{total} with the impinging droplet Weber number. It can be found that for the non-Newtonian droplets, the larger the Weber number of impinging droplet, the faster for the droplets reaches complete mixing. t_{total} and We satisfy the power law, as shown in Eq. (19) ($R^2 = 0.9428$)

$$t_{total} = 6.52 \times 10^5 We^{-0.368} \quad (19)$$

It can be seen from the above that t_{total} can be calculated by Weber number, and will not be too low even the Weber number is high enough. In fact, t_{total} is mainly affected by the duration of molecular diffusion stage which is not always short. We explore the final mixing degree of droplets under different Weber number, as shown in the Fig. 18. The degree of mixing is different from $m_z(t)$. In calculation of the mixing degree, the mixing area can be divided by hundreds of cubic cells. The cell is considered to reach a perfect mixing degree when the number of particles of impinging droplet in it is 40 % –60 % of the total number of particles. The degree of mixing is obtained by counting the ratio of the number of perfect mixing cells to the total number of cells. The mixing degree can also be characterized by the number of particles of ses-

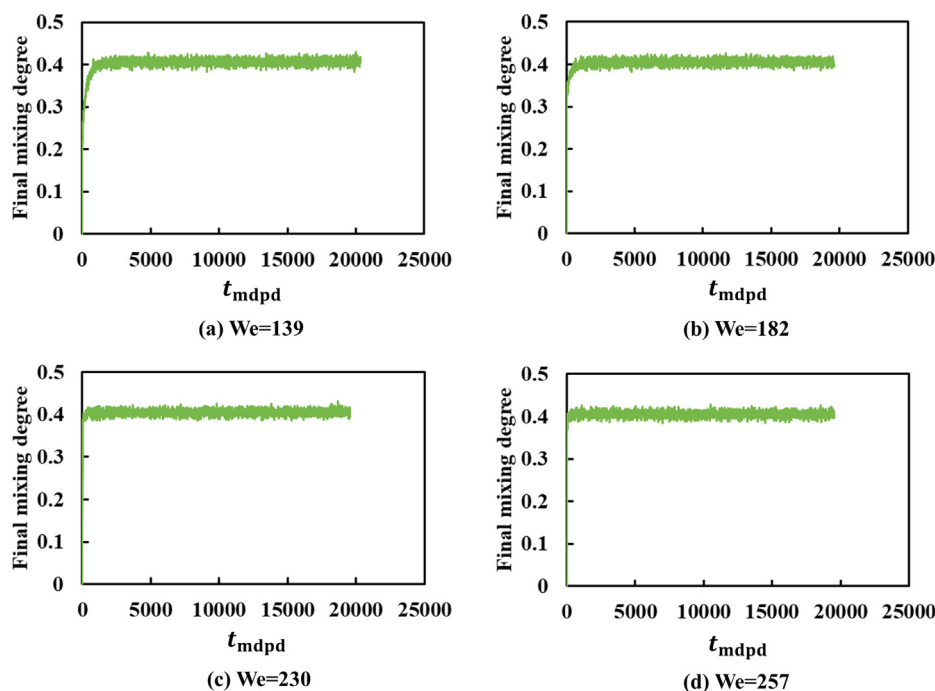


Fig. 18. The degree of mixing when complete mixing is achieved.

sile droplet, and the result obtained is same as that of impinging droplet. The degree of mixing fluctuates around 0.4, and it can also be found that the mixing degree reaches this value when t_{mldp} is around 1000, and the subsequent long mixing process is mainly the molecular diffusion stage.

From Figures 18 and 19, when Weber number of impinging droplet is large, increasing the Weber number has little effect on reducing the mixing time t_{total} . In Section 3.1, we have realized that the droplet morphology does not change with Weber number when Weber number is large. Therefore, in practical applications, an extremely high Weber number of impinging droplets is not necessary when the droplets are non-Newtonian.

4. Conclusion

In this study, the MDPD method was used to numerically simulate the process of non-Newtonian droplets with shear-thinning properties impinging sessile droplets in a “head-to-head” manner. The dynamic behavior and mixing characteristics of droplets in this process were studied, and a special diffusion law of particles between non-Newtonian droplets during the mixing process was revealed. The main conclusions are as follows:

(1) After a single shear-thinning droplet impinges a sessile droplet on a hydrophobic surface, the stable morphology of the droplet is affected by the apparent viscosity and Weber number of the impinging droplet. There is a critical Weber number We_c indicating the critical condition to overcome the resistance of droplet deformation during the impingement.

(2) In the convection-dominated stage, the relationship between t_c and Weber number can be mainly divided into three sections. When the Weber number is lower than 1, the mixing of the two non-Newtonian droplets is poor and dependent on the molecular diffusion. When the Weber number is greater than 205, t_c increases with the increase of the Weber number, which is not beneficial to the rapid mixing. In order to reach a controllable mixing in practice, we should make sure that the Weber number is between 1 and 205.

(3) In the molecular diffusion-dominated mixing stage, due to the difference in diffusion coefficient, particles (type 2) with larger diffusion coefficient will preferentially diffuse from one droplet surface to another, and then from the droplet surface to the inside, while particles (type 1) with smaller diffusion coefficient do not follow the similar behavior.

(4) For the particles with larger diffusion coefficient, total mixing time required for the droplet impingement decreases as the Weber number of the impinging drop increases, and the relationship between total mixing time and Weber number follows the power law. The degree of mixing fluctuates around 0.4 when complete mixing is achieved after droplet impingement.

CRediT authorship contribution statement

Ziqi Li: Investigation, Writing - original draft, Writing - review & editing. **Guina Yi:** Investigation. **Zhengming Gao:** Funding acquisition, Project administration, Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2022.118293>.

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